

Eshaan Vasanthakumar

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EDUCATION

University of Illinois Urbana-Champaign
Bachelor of Science in Mechanical Engineering

Expected May 2028
Urbana-Champaign, IL

EXPERIENCE

Addverb
Research and Development Mechanical Engineering Intern

June 2026 – Present
Noida, India

- Working on **warehouse automation** and **autonomous mobile robots (AMRs)** for intralogistics and material handling.

xTerra Robotics
Robotics Systems Intern

June 2025 – August 2025
Kanpur, India

- Designed and fabricated an aluminum base plate and 3D-printed arm mount for a 6-DOF robot arm, improving **robot integration** safety and mechanical stiffness.
- Developed Python scripts for **coordinate transforms** and **trajectory planning** of a 6-DOF manipulator, enabling smooth joint-space motion in simulation and hardware.
- Tuned **PD/PID controllers** in the ROS control stack and validated tracking repeatability and accuracy against **ISO 9283** robotics performance tests.
- Authored interactive lab-style robotics experiments and documentation to teach **robot kinematics, control, and system integration** using real hardware.

Illinois Robotics in Space
Mechanical Subteam Member

August 2025 – May 2026
Urbana-Champaign, IL

- Designed a linear-actuated bucket dispenser for a Lunabotics rover, enabling **repeatable autonomous regolith dumping** in a planetary robotics context.
- Optimized regolith-handling geometry (clearances, dump angle, lip profile) to reduce **clogging, bridging**, and wear in granular material flow.

PROJECTS

ROS 2 DWA Local Navigation Loop

- Implemented a custom **Dynamic Window Approach (DWA) local planner** in Python for a TurtleBot3 Burger, subscribing to `/odom` and `/scan` and publishing `/cmd_vel` for autonomous mobile robot navigation in Gazebo.
- Sampled (v, ω) within a dynamic window, simulated trajectories, and scored them with goal, heading, obstacle (LaserScan), smoothness, and progress costs, using RViz MarkerArray and logs to debug local planner behavior and **velocity command** outputs.

COURSEWORK

Manufacturing & Automation : ME 451 — industrial robots, manufacturing systems, sensors, actuators, optimization

Materials : ME 330 — material behavior, fatigue, fracture, corrosion, wear, materials selection

Mechanics : ME 270 — statics, force balance, structural load paths, mechanism analysis

Circuits & Electronics : ECE 205/206 — circuit analysis, lab instrumentation, sensors, motor control, feedback, logic

Statistics : STAT 400 — probability, distributions, confidence intervals, hypothesis testing, experimental analysis

Design & CAD : ME 170 — CAD, engineering drawings, assemblies, tolerancing, rapid prototyping, basic kinematics

Thermodynamics : ME 200 — work/heat transfer, control-volume analysis, efficiency, energy systems